PROJECT PLANNING SPECIFICATION

PHASE 4: GLOBAL EV ANALYTICS DASHBOARD IMPLEMENTATION

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Target Engine: Tableau Enterprise
Planning Date: October 2026

KEY MANUFACTURERS ANALYZED	CO2 REDUCTION BASELINE	BATTERY CHEMISTRIES	MAX UNIT SALES RECORDED
12 Global OEMs	13.18 Tons/Yr/Country	LFP, NCA, NMC, Solid-State	> 1.0M Units

1. EXECUTIVE SUMMARY & PROJECT OBJECTIVES

This Project Planning Phase document defines the technical execution, data integration strategy, architecture, and schedule for deploying the **Global Electric Vehicle (EV) Performance and Market Analytics Suite**. Built upon verified Tableau production templates, the platform analyzes manufacturing output, drivetrain distributions (AWD, FWD, RWD), battery degradation kinetics, CO2 offset metrics, and pricing trends across major vehicle categories.

2. VISUAL ANALYTICS & DASHBOARD COMPONENT SPECIFICATION

Dashboard 1: Macro Production & Environmental Impact

Includes donut visualizations of annual sales by battery chemistry, manufacturer-level battery degradation plots (ranging from VoltaAuto to AmpereMobility), country-level CO2 offsets (consistent ~13.18 tons/year across key manufacturing hubs), and scatter correlation of Sales vs 0-100 km/h acceleration.

Dashboard 2: Market Segmentation & Pricing Trends

Integrates a Sankey Drivetrain Flow (mapping China, France, Germany, India, Japan, UK, USA, Vietnam to AWD/FWD/RWD), a Category Market Share Treemap (MicroCity, Full-size SUV, Hatchback), a Battery Performance Cross-Tab Matrix, and multi-year Average Price Trends ($10K-$100K+).

Data Ingestion Requirements

Automated ETL pipeline connecting OEM production feeds, vehicle telematics (battery degradation rates), customer ratings (scale 1.52 to 5.00), and historical market pricing from 2017 through 2025.

Governance & Access Control

Row-level security (RLS) configured in Tableau Server for regional analysts, ensuring strict data privacy for proprietary OEM performance indices and battery test metrics.

3. WORK BREAKDOWN STRUCTURE (WBS) & MILESTONES

The implementation plan is structured into five distinct phases over a 10-week execution timeline. Deliverables are validated against raw data schema standards and functional dashboard specifications.

WBS ID	Phase / Deliverable Name	Key Tasks & Technical Scope	Duration	Status
1.0	Data Architecture & ETL Pipeline	Aggregate OEM telematics, battery degradation feeds, drivetrain specs, and CO2 indices.	Weeks 1-2	COMPLETED
2.0	Data Modeling & Schema Design	Establish star-schema data warehouse for sales metrics, battery type matrices, and vehicle categories.	Weeks 3-4	COMPLETED
3.1	Dashboard 1 Visualization	Build Donut Sales chart, Degradation scatter/strip plots, and CO2 manufacturing bar charts.	Weeks 5-6	IN PROGRESS
3.2	Dashboard 2 Visualization	Construct Drivetrain Flow Sankey, Category Market Share Treemap, and Price Trend lines.	Weeks 6-7	IN PROGRESS
4.0	Quality Assurance & Validation	Validate acceleration ranges (0-16s), null filtering (>2K records), and pricing calculations.	Weeks 8-9	PENDING
5.0	Deployment & Server Publishing	Configure Tableau Server permissions, embedded tooltips, and interactive filter parameters.	Week 10	PENDING

4. DETAILED DASHBOARD FEATURE & DATA ELEMENT MAPPING

Drivetrain Distribution Analytics

Captures production allocation across AWD, FWD, and RWD modes. Ensures multi-country data harmonization for manufacturing powerhouses (China, India, USA, Germany, Japan, South Korea, Sweden, UK, Vietnam, France).

Battery Chemistry Matrix (Sheet 10)

Cross-tabulation evaluating LFP, NCA, NMC, Sodium-Ion, and Solid-State across vehicle segments: Compact, Crossover, Electric Truck, Full-size SUV, Hatchback, Luxury, MicroCity Car, Pickup, Sedan.

5. RISK MANAGEMENT & MITIGATION MATRIX

Key technical and operational risks identified during project planning, along with assigned mitigation strategies and ownership.

Risk ID	Risk Event / Description	Impact	Likelihood	Mitigation Strategy
RSK-01	Data missingness in acceleration telematics (>2K null values recorded)	Medium	High	Implement data imputation rules and explicit 'Unknown' filter categories in Tableau.
RSK-02	High rendering latency for Sankey drivetrain flow diagram	High	Medium	Pre-calculate curve coordinates in SQL/Python layer prior to loading into Tableau.
RSK-03	Discrepancies in battery degradation testing standards across OEMs	High	Low	Normalize degradation values to percentage degradation per 10k miles driven.
RSK-04	Price trend distortion due to currency fluctuations (2017–2025)	Medium	Medium	Standardize all price points to USD equivalent using annual average spot rates.

6. RESOURCE ALLOCATION & SIGN-OFF SCHEDULE

Project Team Roles

- **Lead Data Engineer:** ETL & Data Warehouse
- **Tableau Architect:** Dashboard Development
- **EV Domain Expert:** Telematics & Battery QA
- **Project Manager:** Schedule & Stakeholder Align

Project Sign-Off Criteria

1. Dashboard render time < 2.5 seconds.
2. 100% data reconciliation between raw telematics and Tableau views.
3. Final UAT sign-off from Regional Analytics Directors.